

Equity Yield Rate and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Yield Rate (EYR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EYR achieves an annualized gross (net) Sharpe ratio of 0.55 (0.52), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (30) bps/month with a t-statistic of 2.68 (3.29), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly)) is 20 bps/month with a t-statistic of 2.32.

1 Introduction

Asset prices reflect firms' production decisions and the frictions they face when adjusting capital and investment. In competitive markets, firms equate marginal products of capital to their costs, creating a fundamental link between production efficiency and expected returns. Despite extensive research on production-based asset pricing models, significant cross-sectional variation in stock returns remains unexplained by standard measures of firms' investment opportunities and capital productivity. The equity yield rate (EYR), defined as the ratio of operating cash flow to market equity, offers a direct measure of the productivity of shareholders' capital investment. This signal captures the current cash return generated per dollar of equity market value, potentially proxying for firms' marginal efficiency of capital and their exposure to productivity shocks that drive time-varying risk premia in production economies.

In a production-based framework, firms facing higher adjustment costs or binding financing constraints exhibit lower investment despite profitable opportunities, leading to higher marginal products of capital [Cochrane \(1991\)](#). The equity yield rate directly measures this marginal productivity by scaling operating cash flows by the market's valuation of equity capital. When firms face convex adjustment costs, those with higher EYR have accumulated productive assets that generate substantial cash flows relative to their market values, indicating either underinvestment due to frictions or exposure to systematic productivity risk [Zhang \(2005\)](#). These high-EYR firms earn higher expected returns as compensation for their exposure to aggregate productivity shocks that affect the economy's stochastic discount factor.

The production-based interpretation of EYR aligns with models where cross-sectional return variation arises from heterogeneous exposures to investment-specific technology shocks [Papanikolaou \(2011\)](#). Firms with high equity yields operate with capital that generates substantial current cash flows, suggesting mature, productivity-

sensitive assets in place. These firms face greater risks during economic downturns when productivity shocks reduce the value of existing capital, requiring higher expected returns in equilibrium. The cash flow component of EYR captures the realized productivity of capital, while the market equity denominator reflects investors' forward-looking assessments of growth options and adjustment flexibility.

Furthermore, EYR connects to the q -theory of investment where firms' investment-to-capital ratios depend on marginal q , the shadow value of installed capital [Liu et al. \(2009\)](#). High-EYR firms exhibit characteristics consistent with high marginal products of capital relative to average q (market-to-book ratios), suggesting these firms face either significant adjustment frictions or systematic risks that prevent them from scaling operations despite profitable opportunities. The resulting cross-sectional variation in expected returns reflects compensation for bearing these production-related risks, particularly exposure to shocks affecting the productivity of capital in place versus growth options.

We find strong evidence that equity yield rate predicts cross-sectional stock returns. A value-weighted long-short trading strategy based on EYR achieves an annualized gross Sharpe ratio of 0.55 and a net Sharpe ratio of 0.52 after accounting for transaction costs. The strategy delivers a monthly average excess return of 54 basis points with a t -statistic of 3.32. Most notably, the EYR strategy generates significant abnormal returns relative to standard factor models, with monthly alphas ranging from 25 to 77 basis points depending on the factor model specification. The alpha relative to the Fama-French six-factor model equals 25 basis points per month with a t -statistic of 2.68, demonstrating that EYR captures return variation beyond established risk factors including market, size, value, profitability, investment, and momentum.

The economic significance of the EYR effect remains substantial across different portfolio construction methodologies and size groups. Among the largest stocks—those

with market capitalizations above the 80th NYSE percentile—the EYR strategy generates an average return of 46 basis points per month with a t-statistic of 3.15. The strategy’s performance proves robust to transaction costs, with net returns remaining statistically and economically significant across all portfolio construction methods tested. The net alpha relative to the six-factor model equals 30 basis points per month with a t-statistic of 3.29, indicating that EYR significantly expands the mean-variance efficient frontier even after accounting for implementation costs.

Controlling for closely related anomalies further validates the distinct information content of EYR. When we simultaneously control for the six most closely related predictors—operating profits to book equity, net income to book equity, analyst earnings per share, net external financing, idiosyncratic risk, and quarterly return on assets—plus the six Fama-French factors, the EYR strategy maintains an alpha of 20 basis points per month with a t-statistic of 2.32. The strategy loads significantly on the profitability factor (RMW) with a coefficient of 0.76, consistent with EYR capturing firms’ productive efficiency, yet the remaining alpha indicates that EYR contains unique information about expected returns beyond standard profitability measures.

Our study contributes to the production-based asset pricing literature by establishing equity yield rate as a powerful predictor of cross-sectional returns that captures firms’ marginal productivity of capital. Unlike [Fama and French \(2015\)](#) profitability factor (RMW) which scales earnings by book equity, EYR uses operating cash flows scaled by market equity, providing a more direct measure of the current productivity of shareholders’ invested capital. While [Novy-Marx \(2013\)](#) demonstrates that gross profitability predicts returns, our EYR measure focuses on cash generation rather than accounting profits, avoiding potential distortions from accruals and better capturing the timing of actual cash flows that matter for production decisions. Relative to [Hou et al. \(2015\)](#) investment-based factors, EYR provides

a complementary perspective by measuring the output efficiency of existing capital rather than the rate of new capital formation.

Methodologically, we employ the comprehensive testing framework of [Novy-Marx and Velikov \(2023b\)](#) to establish the robustness and economic significance of the EYR anomaly. This protocol ensures that our results are not artifacts of particular portfolio construction choices or vulnerable to implementation costs that plague many documented anomalies. The EYR strategy’s gross Sharpe ratio of 0.55 places it in the 93rd percentile among 212 anomalies in the factor zoo, while its net Sharpe ratio of 0.52 ranks in the 99th percentile, demonstrating exceptional performance after accounting for transaction costs. These results survive controlling for the most closely related anomalies from [Chen and Zimmermann \(2022\)](#) comprehensive dataset, confirming that EYR contains distinct information about the cross-section of expected returns.

The broader implications of our findings extend beyond identifying a new return predictor to validating production-based theories of asset pricing. The strong performance of EYR, particularly its significant loading on profitability factors while maintaining substantial alpha, suggests that markets price productivity risk beyond what standard factors capture. The persistence of the EYR premium after controlling for transaction costs and related anomalies indicates that frictions in production and capital adjustment create lasting differences in expected returns. Our results support models where heterogeneous exposure to productivity shocks and investment frictions generate cross-sectional return variation, providing empirical validation for production-based asset pricing theories that link firm characteristics to systematic risk exposures through the lens of optimal production decisions.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of cash flows and equity values across firms. Equity Yield Rate signal requires two key variables from COMPUSTAT. Operating activities net cash flow (OANCF) represents the cash generated from a firm’s core business operations, calculated using the indirect method that adjusts net income for non-cash charges and changes in working capital. This measure captures the actual cash a company generates from its primary business activities, excluding cash flows from investing and financing activities. Equity liquidation value (CEQL) represents the book value of common equity plus any preferred stock liquidation value, providing a measure of the residual claim that equity holders have on the firm’s assets after all liabilities are satisfied. We construct the Equity Yield Rate signal by dividing operating activities net cash flow (OANCF) by equity liquidation value (CEQL) for each firm-year observation. This ratio captures the cash-generating efficiency of equity capital, measuring how much operating cash flow a firm generates per dollar of equity liquidation value. We calculate the signal using fiscal year-end values to ensure alignment between the flow variable (OANCF) and the stock variable (CEQL). To maintain data quality, we require non-missing values for both OANCF and CEQL, and exclude observations where CEQL equals zero to avoid undefined ratios. The resulting signal provides a cash-based measure of return on equity that complements traditional accrual-based profitability metrics.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EYR signal. Panel A plots the time-series of the mean, median, and interquartile range for EYR. On average, the cross-sectional mean (median) EYR is 0.08 (0.14) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input EYR data. The signal’s interquartile range spans -0.24 to 0.30. Panel B of Figure 1 plots the time-series of the coverage of the EYR signal for the CRSP universe. On average, the EYR signal is available for 7.37% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does EYR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EYR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EYR portfolio and sells the low EYR portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EYR strategy earns an average return of 0.54% per month with a t-statistic of 3.32. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.25% to 0.77% per month and have t-statistics exceeding 2.68 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.76,

with a t-statistic of 18.11 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 525 stocks and an average market capitalization of at least \$1,720 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 38 bps/month with a t-statistics of 2.48. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 19-51bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EYR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the EYR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EYR, as well as average returns and alphas for long/short trading EYR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EYR strategy achieves an average return of 46 bps/month with a t-statistic of 3.15. Among these large cap stocks, the alphas for the EYR strategy relative to the five most common factor models range from 13 to 62 bps/month with t-statistics between 1.14 and 4.57.

5 How does EYR perform relative to the zoo?

Figure 2 puts the performance of EYR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EYR strategy falls in the distribution. The EYR strategy’s gross (net) Sharpe ratio of 0.55 (0.52) is greater than 93% (99%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EYR strategy (red line).² Ignoring trading costs, a \$1 invested in the EYR strategy would have yielded \$7.31 which ranks the EYR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EYR strategy would have yielded \$6.42 which ranks the EYR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EYR relative to those. Panel A shows that the EYR strategy gross alphas fall between the 74 and 95 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EYR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EYR ranks between the 93 and 99 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EYR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EYR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EYR or at least to weaken the power EYR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EYR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EYR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EYR}EYR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EYR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EYR. Stocks are finally grouped into five EYR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EYR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EYR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EYR signal in these Fama-MacBeth regressions exceed 0.06, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on EYR is 2.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the EYR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EYR strategy earns alphas that range from 21-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.28, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EYR trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.32.

7 Does EYR add relative to the whole zoo?

Finally, we can ask how much adding EYR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the EYR signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EYR is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy that includes EYR grows to \$43.54.

8 Conclusion

This study provides compelling evidence for the predictive power of the Equity Yield Rate (EYR) signal in cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EYR represents a robust and economically significant predictor of future equity returns. The value-weighted long/short strategy based on EYR delivers impressive risk-adjusted returns, with annualized Sharpe ratios of 0.55 (gross) and 0.52 (net), indicating strong performance even after accounting for transaction costs. The signal’s economic significance is further underscored by its ability to generate statistically significant alpha relative to the Fama-French five-factor model plus momentum, with monthly abnormal returns of 25-30 basis points and t-statistics exceeding conventional significance thresholds.

Perhaps most notably, EYR maintains its predictive power even when controlling for twelve factors, including the six standard risk factors and six closely related strategies from the factor zoo. The persistence of a 20 basis points monthly alpha with a t-statistic of 2.32 in this stringent specification suggests that EYR captures unique information about expected returns not subsumed by existing factors or similar profitability and financing measures. This finding has important practical implications

for both asset managers and academic researchers, as it identifies a viable strategy that can enhance portfolio performance while contributing to our understanding of the cross-section of expected returns.

From a practical perspective, the relatively modest degradation from gross to net returns suggests that implementation costs do not materially erode the strategy’s profitability, making it potentially attractive for institutional investors. The signal’s robustness across multiple factor models also indicates that it may serve as a valuable addition to multi-factor investment strategies.

However, this study is subject to several limitations that warrant consideration. First, while we employ state-of-the-art methodologies to address data-mining concerns, the proliferation of factors in the literature necessitates continued vigilance regarding out-of-sample performance. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying EYR’s predictive power deserve further theoretical and empirical investigation to better understand why this signal contains information incremental to established factors.

Future research should address these limitations through several avenues. International replication studies would help establish the global validity of the EYR signal. Time-varying analysis could reveal whether the signal’s effectiveness varies across different market regimes or economic cycles. Additionally, investigating the interaction between EYR and firm characteristics such as size, industry, or financial constraints could provide deeper insights into when and where the signal is most effective. Finally, exploring the theoretical foundations of why EYR predicts returns—whether through risk-based or behavioral channels—would enhance our understanding of asset pricing dynamics and potentially improve strategy implementation. As the factor zoo continues to expand, signals like EYR that demonstrate robust performance across stringent tests remain valuable contributions to both academic literature and

investment practice.

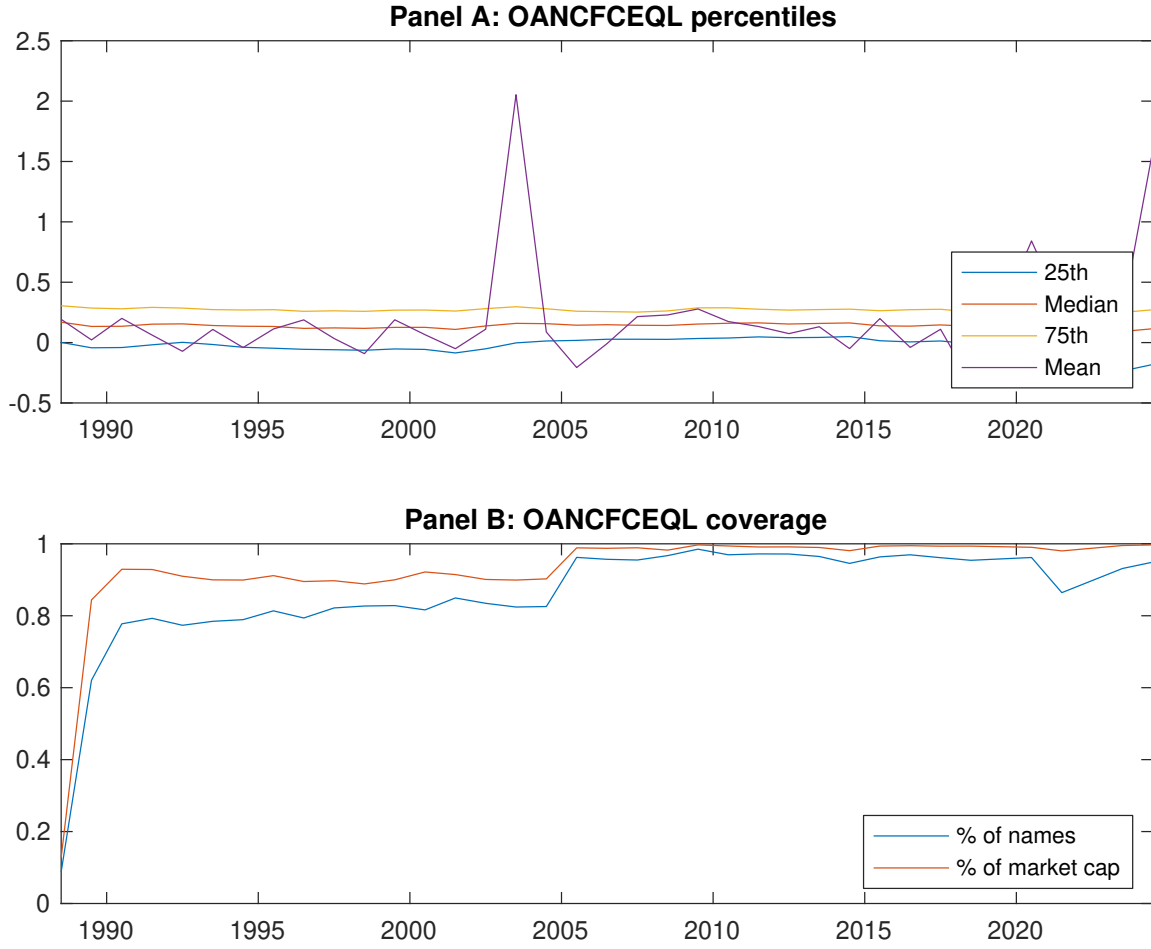


Figure 1: Times series of EYR percentiles and coverage. This figure plots descriptive statistics for EYR. Panel A shows cross-sectional percentiles of EYR over the sample. Panel B plots the monthly coverage of EYR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EYR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on EYR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.39 [1.39]	0.57 [2.49]	0.69 [3.19]	0.78 [3.92]	0.93 [4.54]	0.54 [3.32]
α_{CAPM}	-0.54 [-4.93]	-0.20 [-3.02]	-0.04 [-0.63]	0.10 [1.86]	0.24 [4.04]	0.77 [5.16]
α_{FF3}	-0.49 [-5.63]	-0.22 [-3.57]	-0.06 [-0.95]	0.10 [1.84]	0.23 [4.46]	0.72 [6.07]
α_{FF4}	-0.44 [-4.99]	-0.19 [-3.08]	-0.04 [-0.67]	0.13 [2.47]	0.19 [3.75]	0.63 [5.33]
α_{FF5}	-0.20 [-2.84]	-0.10 [-1.77]	-0.06 [-1.00]	0.04 [0.81]	0.10 [2.15]	0.30 [3.26]
α_{FF6}	-0.17 [-2.40]	-0.09 [-1.45]	-0.05 [-0.74]	0.07 [1.37]	0.08 [1.63]	0.25 [2.68]
Panel B: Fama and French (2018) 6-factor model loadings for EYR-sorted portfolios						
β_{MKT}	1.09 [60.90]	0.99 [66.87]	0.99 [62.24]	0.93 [68.43]	1.02 [86.73]	-0.07 [-2.89]
β_{SMB}	0.26 [10.22]	0.09 [4.30]	0.09 [3.80]	-0.06 [-3.19]	-0.11 [-6.31]	-0.36 [-11.12]
β_{HML}	0.09 [3.00]	0.19 [7.39]	0.08 [2.74]	-0.07 [-2.73]	-0.10 [-5.05]	-0.20 [-4.89]
β_{RMW}	-0.53 [-16.25]	-0.20 [-7.55]	-0.03 [-1.10]	0.13 [5.32]	0.23 [10.88]	0.76 [18.11]
β_{CMA}	-0.22 [-4.93]	-0.09 [-2.25]	0.10 [2.46]	0.02 [0.60]	0.09 [2.97]	0.31 [5.32]
β_{UMD}	-0.05 [-3.12]	-0.03 [-2.17]	-0.03 [-1.84]	-0.05 [-3.97]	0.04 [3.73]	0.09 [4.31]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1566	797	590	525	583	
me (\$10 ⁶)	1720	2163	2645	4552	5284	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EYR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.54 [3.32]	0.77 [5.16]	0.72 [6.07]	0.63 [5.33]	0.30 [3.26]	0.25 [2.68]
Quintile	NYSE	EW	0.38 [2.48]	0.46 [2.98]	0.34 [2.84]	0.24 [2.02]	0.02 [0.22]	-0.05 [-0.51]
Quintile	Name	VW	0.43 [2.30]	0.68 [3.90]	0.61 [4.35]	0.58 [4.03]	0.28 [2.23]	0.26 [2.06]
Quintile	Cap	VW	0.43 [3.00]	0.60 [4.33]	0.57 [4.92]	0.48 [4.17]	0.17 [1.88]	0.12 [1.29]
Decile	NYSE	VW	0.52 [2.85]	0.76 [4.39]	0.68 [4.67]	0.63 [4.29]	0.24 [1.96]	0.22 [1.78]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.51 [3.16]	0.75 [4.99]	0.69 [5.79]	0.63 [5.38]	0.34 [3.67]	0.30 [3.29]
Quintile	NYSE	EW	0.19 [1.23]	0.28 [1.73]	0.17 [1.36]	0.11 [0.92]		
Quintile	Name	VW	0.40 [2.13]	0.66 [3.76]	0.58 [4.13]	0.56 [3.95]	0.30 [2.35]	0.29 [2.23]
Quintile	Cap	VW	0.41 [2.87]	0.57 [4.10]	0.53 [4.57]	0.47 [4.14]	0.21 [2.24]	0.17 [1.83]
Decile	NYSE	VW	0.49 [2.67]	0.72 [4.12]	0.64 [4.35]	0.61 [4.12]	0.26 [2.08]	0.24 [1.94]

Table 3: Conditional sort on size and EYR

This table presents results for conditional double sorts on size and EYR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EYR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EYR and short stocks with low EYR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EYR Quintiles					EYR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.22 [0.49]	0.44 [1.22]	0.64 [2.18]	0.86 [3.25]	1.06 [3.41]	0.85 [3.20]	1.09 [4.18]	0.89 [4.22]	0.85 [3.94]	0.45 [2.38]	0.42 [2.20]
	(2)	0.33 [0.85]	0.56 [1.83]	0.77 [2.89]	0.93 [3.59]	1.03 [3.65]	0.70 [3.23]	0.94 [4.50]	0.77 [4.72]	0.75 [4.54]	0.33 [2.46]	0.32 [2.41]
	(3)	0.33 [0.96]	0.67 [2.37]	0.75 [3.06]	0.96 [3.98]	1.03 [3.88]	0.70 [3.80]	0.90 [4.97]	0.75 [5.26]	0.70 [4.83]	0.32 [2.68]	0.29 [2.44]
	(4)	0.64 [2.09]	0.68 [2.73]	0.82 [3.62]	0.88 [3.84]	0.92 [3.77]	0.28 [1.84]	0.45 [3.07]	0.35 [2.78]	0.31 [2.38]	0.01 [0.09]	-0.02 [-0.15]
	(5)	0.42 [1.67]	0.65 [2.89]	0.62 [3.07]	0.95 [4.69]	0.87 [4.20]	0.46 [3.15]	0.60 [4.25]	0.62 [4.57]	0.49 [3.70]	0.21 [1.84]	0.13 [1.14]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EYR Quintiles					EYR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	439	451	455	457	447	39	46	54	63	60	
	(2)	132	132	133	133	132	88	91	94	97	96	
	(3)	89	90	90	90	90	151	158	160	164	165	
	(4)	74	74	74	75	74	324	330	345	347	350	
(5)	66	66	66	66	66	2067	1875	2295	3256	3649		

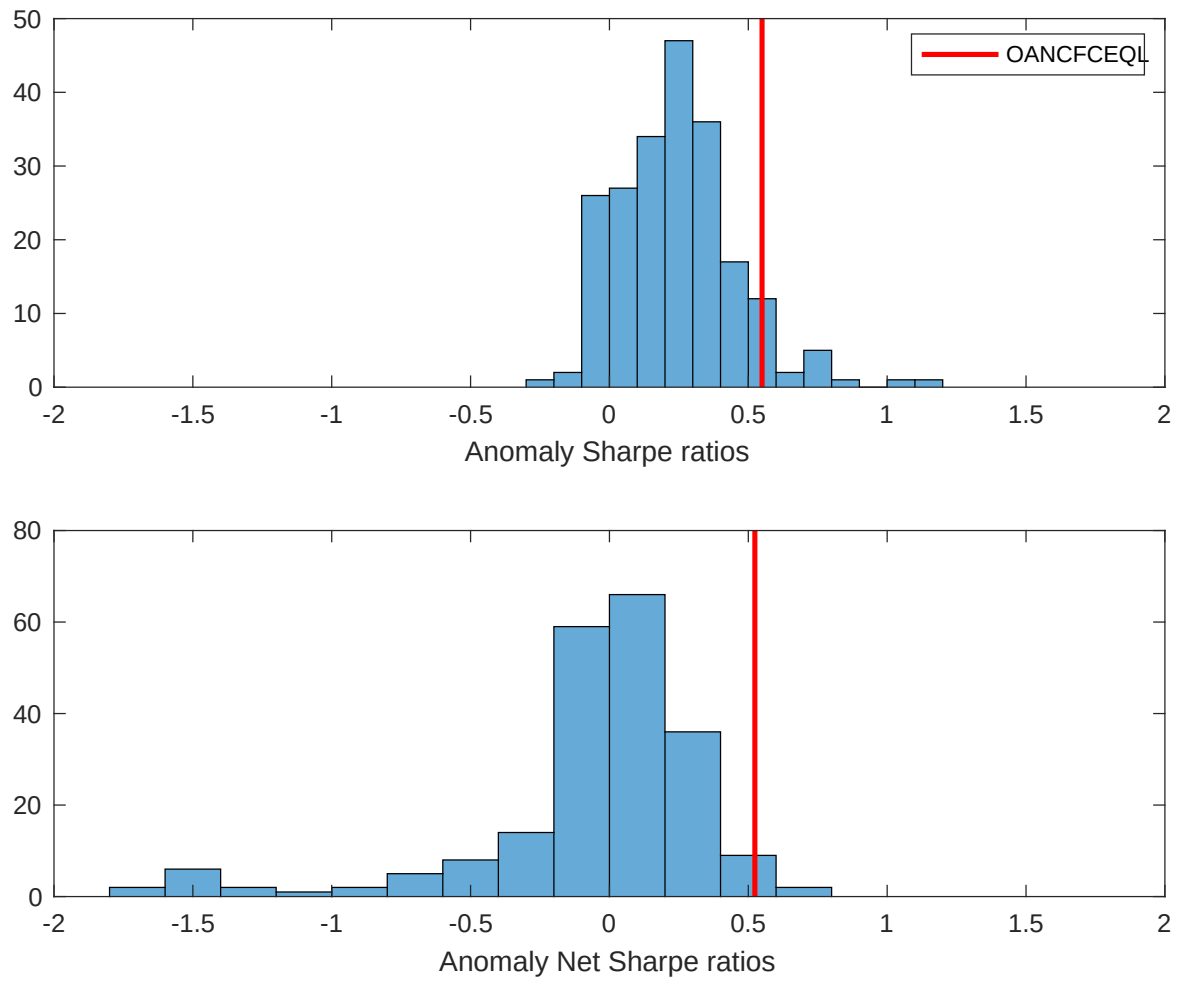


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EYR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

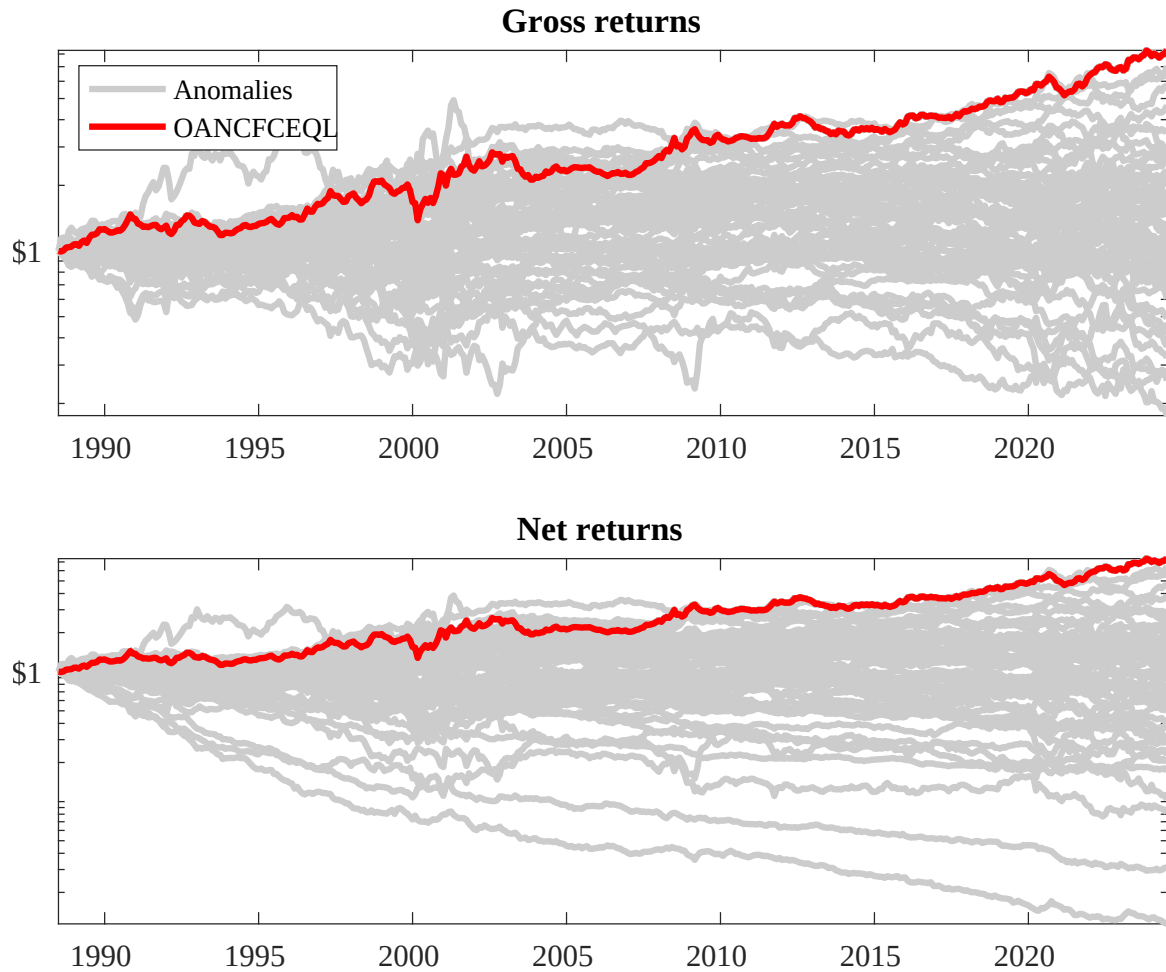
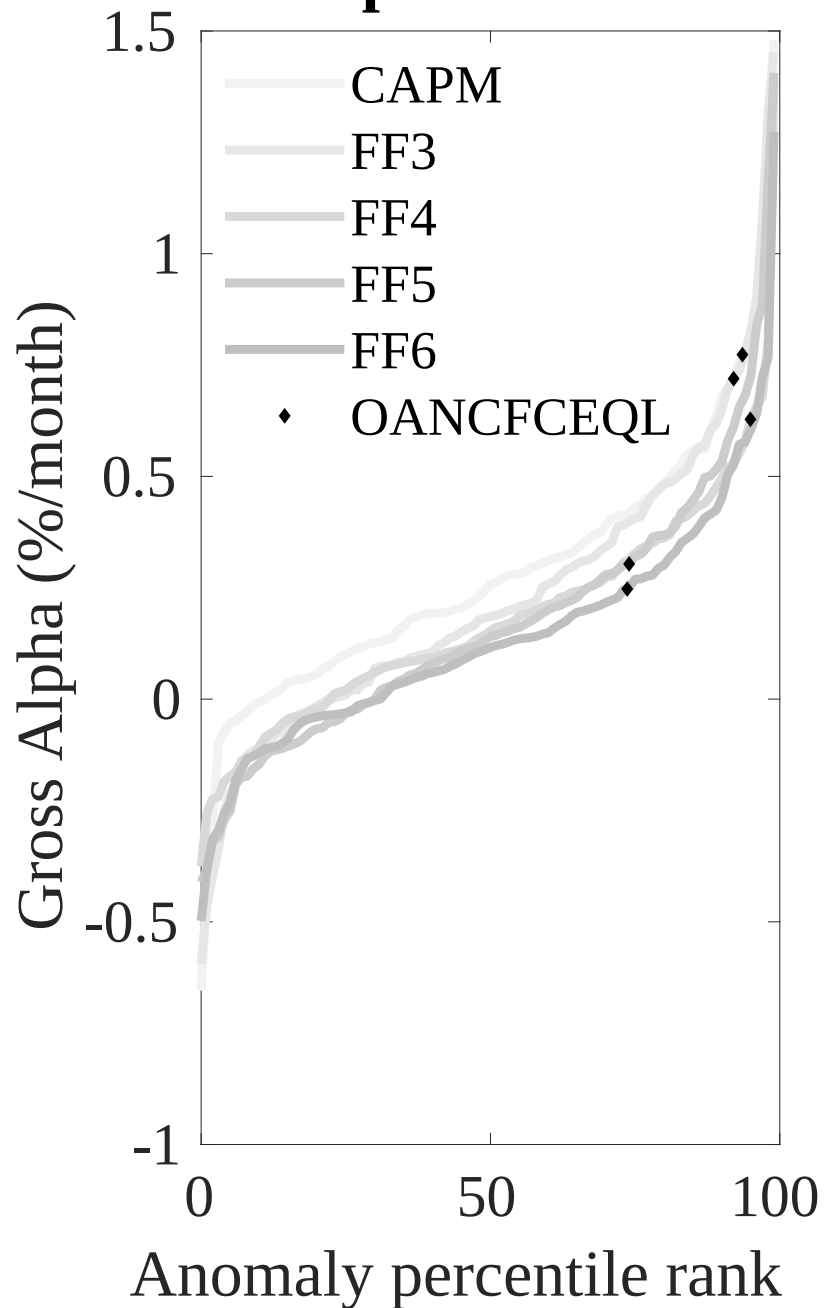


Figure 3: Dollar invested.

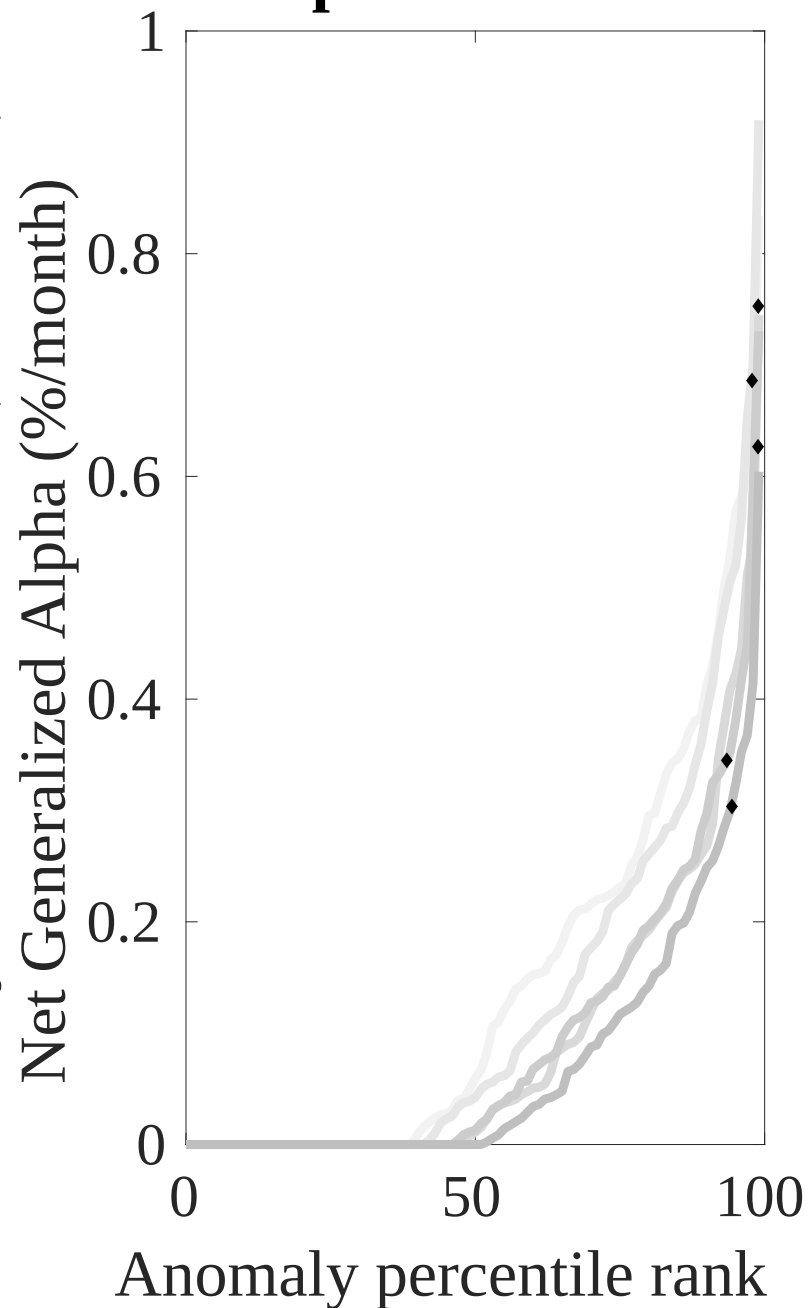
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EYR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



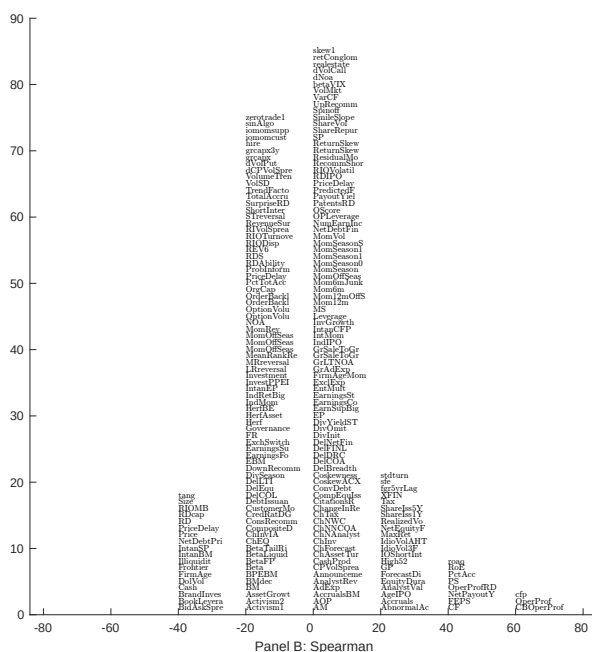
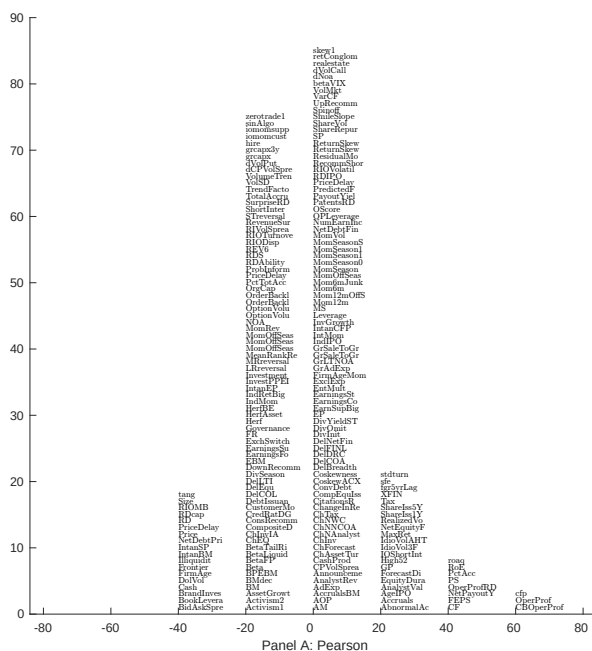


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EYR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

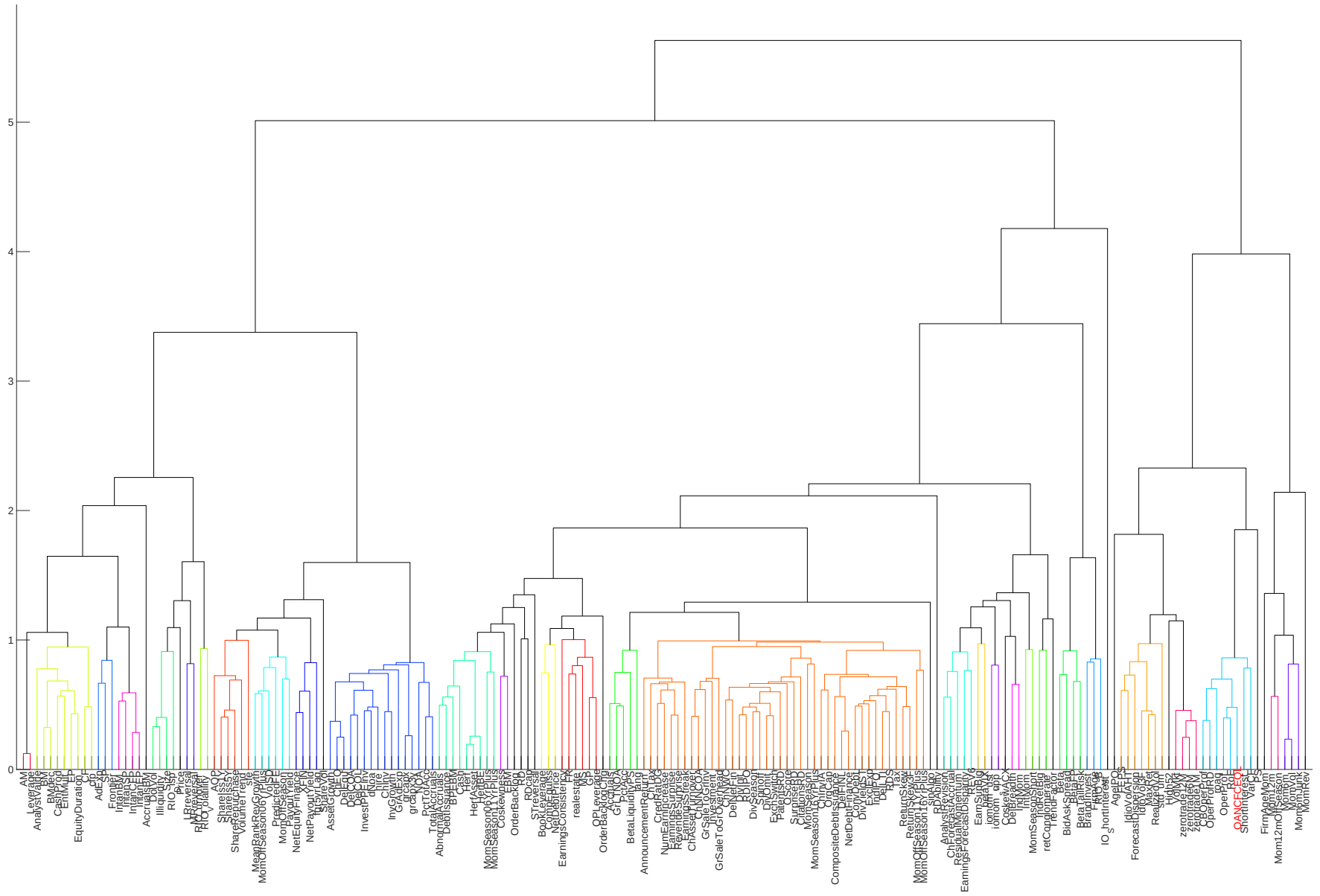


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

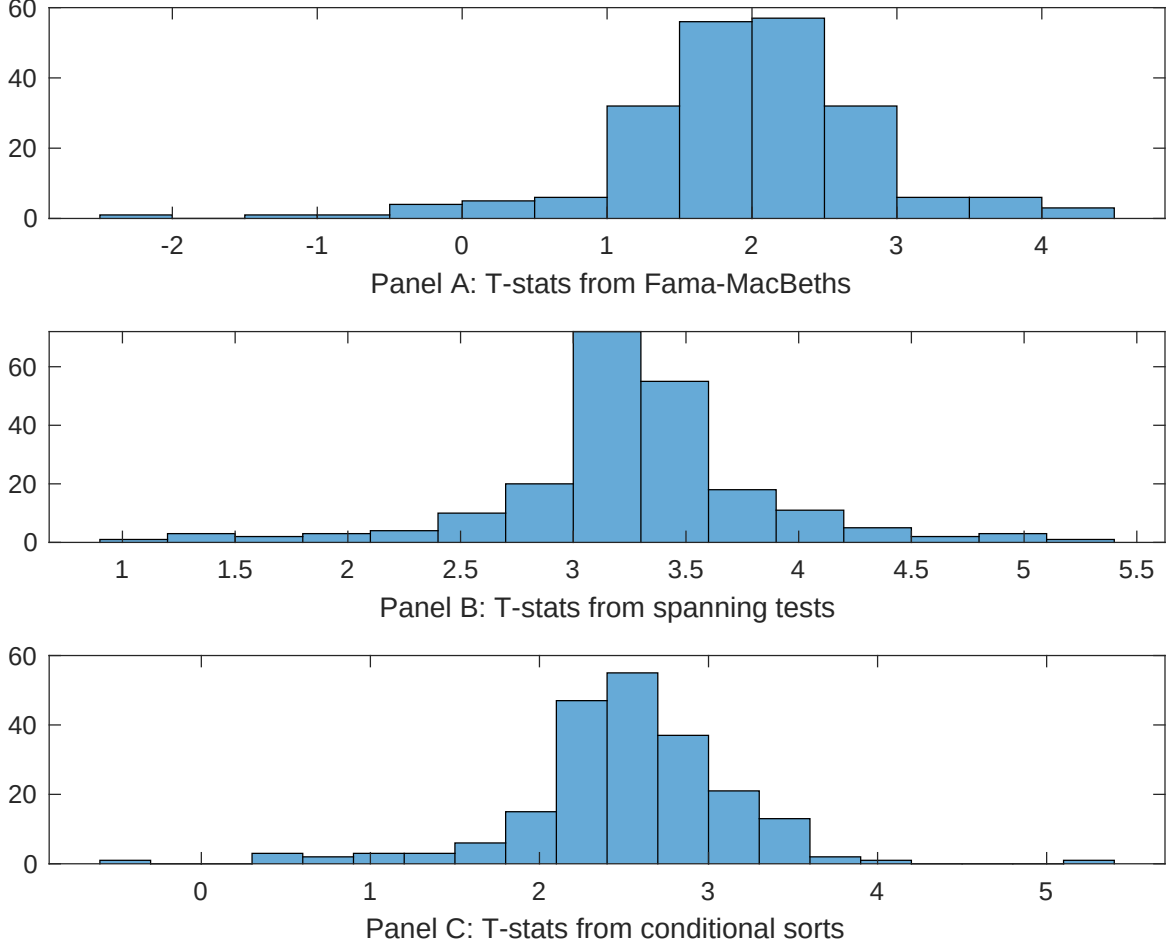


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EYR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EYR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EYR} EYR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EYR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EYR. Stocks are finally grouped into five EYR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EYR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EYR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EYR} EYR_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.89 [3.02]	0.10 [3.49]	0.87 [2.54]	0.12 [4.27]	0.11 [4.78]	0.11 [4.00]	0.11 [5.08]
EYR	0.16 [2.62]	0.20 [2.80]	0.18 [2.96]	0.43 [0.06]	0.18 [3.71]	0.51 [0.97]	0.17 [2.62]
Anomaly 1	0.25 [2.79]						-0.21 [-0.28]
Anomaly 2		0.54 [0.01]					-0.14 [-2.61]
Anomaly 3			0.51 [0.74]				-0.14 [-0.48]
Anomaly 4				0.17 [5.67]			0.11 [4.38]
Anomaly 5					0.76 [1.25]		0.60 [0.75]
Anomaly 6						0.42 [2.69]	0.36 [3.16]
# months	427	432	427	432	427	427	427
$\bar{R}^2(\%)$	0	0	2	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EYR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EYR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are operating profits / book equity, net income / book equity, Analyst earnings per share, Net external financing, Idiosyncratic risk (AHT), Return on assets (qtrly). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.22 [2.51]	0.24 [2.59]	0.25 [2.70]	0.21 [2.28]	0.25 [2.70]	0.24 [2.57]	0.20 [2.32]
Anomaly 1	35.23 [8.57]						36.77 [7.53]
Anomaly 2		21.12 [3.80]					-3.97 [-0.63]
Anomaly 3			-2.83 [-0.80]				-17.49 [-4.23]
Anomaly 4				20.84 [4.68]			11.51 [2.37]
Anomaly 5					3.42 [1.15]		4.47 [1.34]
Anomaly 6						6.10 [1.48]	4.58 [1.10]
mkt	-2.52 [-1.15]	-3.49 [-1.45]	-7.02 [-2.87]	-3.39 [-1.44]	-4.97 [-1.89]	-5.52 [-2.31]	-2.61 [-1.07]
smb	-28.41 [-8.80]	-30.16 [-8.10]	-38.65 [-9.75]	-30.01 [-8.42]	-34.03 [-8.14]	-35.71 [-10.43]	-31.39 [-7.86]
hml	-14.09 [-3.72]	-17.30 [-4.29]	-18.62 [-4.55]	-16.93 [-4.24]	-19.97 [-4.83]	-17.05 [-3.99]	-9.97 [-2.42]
rmw	44.06 [8.28]	56.88 [8.95]	78.43 [13.66]	62.80 [12.86]	71.68 [13.65]	69.76 [12.40]	49.71 [7.48]
cma	23.79 [4.30]	34.20 [5.86]	32.59 [5.43]	18.42 [2.86]	30.44 [5.07]	31.37 [5.32]	18.95 [3.04]
umd	4.89 [2.49]	8.01 [3.92]	9.82 [4.48]	8.52 [4.23]	8.43 [3.89]	7.65 [3.31]	6.15 [2.78]
# months	428	432	428	432	428	428	428
$\bar{R}^2(\%)$	76	72	72	73	72	72	77

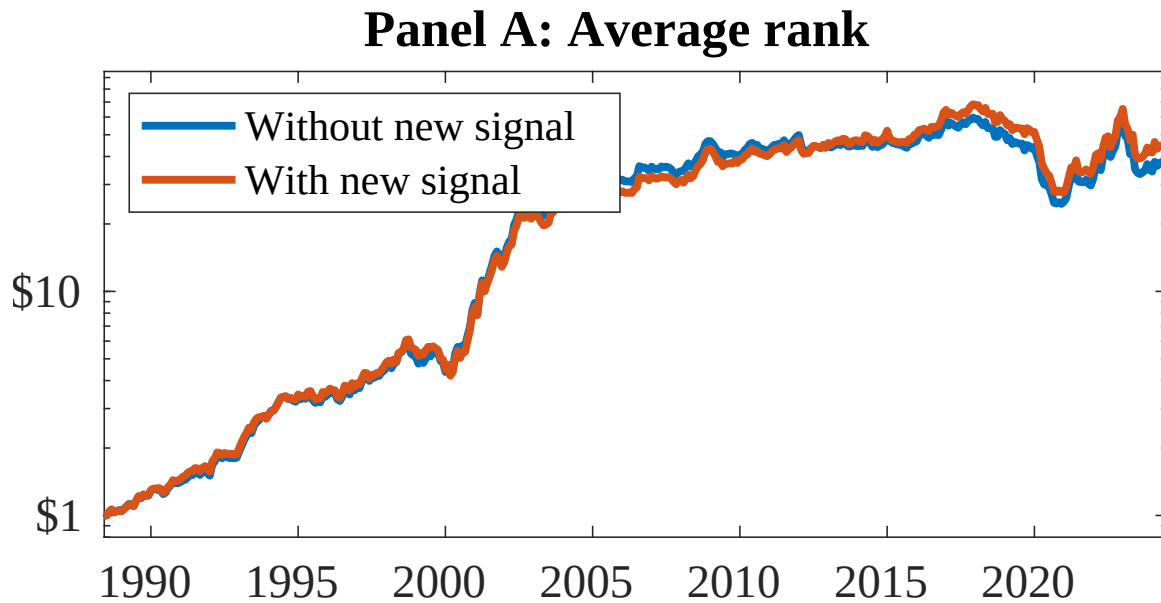


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as EYR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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